Industrial surface defect segmentation

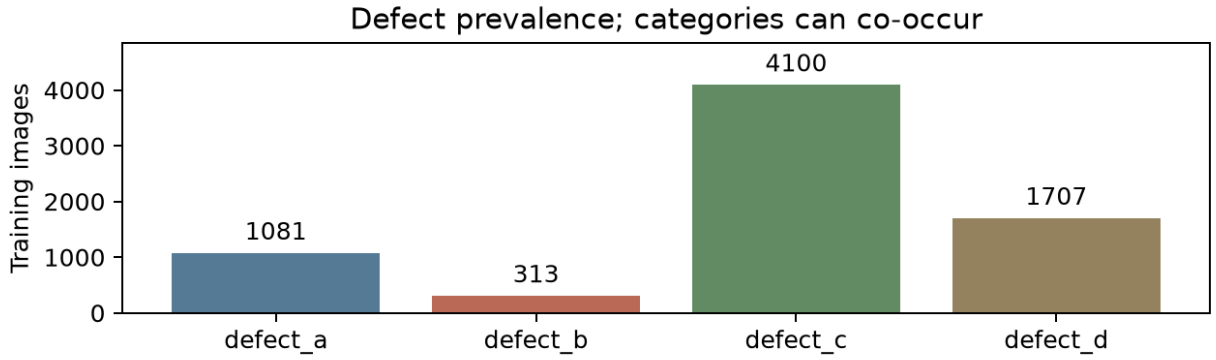
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Abstract and objective

This project predicts four pixel-level defect masks for grayscale industrial strips. A scratch U-Net, an ImageNet-initialized ResNet34 U-Net and an EfficientNet-B0 feature pyramid model were compared on a common image split. The selected inference variant was **equal-weight ensemble**, using Scratch U-Net, ResNet34 U-Net, EfficientNet-B0 FPN. Its executed Kaggle submission scored **0.67008**, with public rank **147 of 1140** in the saved leaderboard snapshot. Validation, public scoring and inference cost are reported separately.

Dataset and exploration

The training data contains 11,393 images and 45,572 image/category annotations. Every strip is 128 x 800; the aspect ratio is 6.25. There are 3,000 test images and 12,000 submission rows. 5,516 training images are clean, and 84.2% of image/category targets are empty. Defect B is rare, so raw pixel accuracy or a naive empty-mask baseline can conceal poor defect recall.



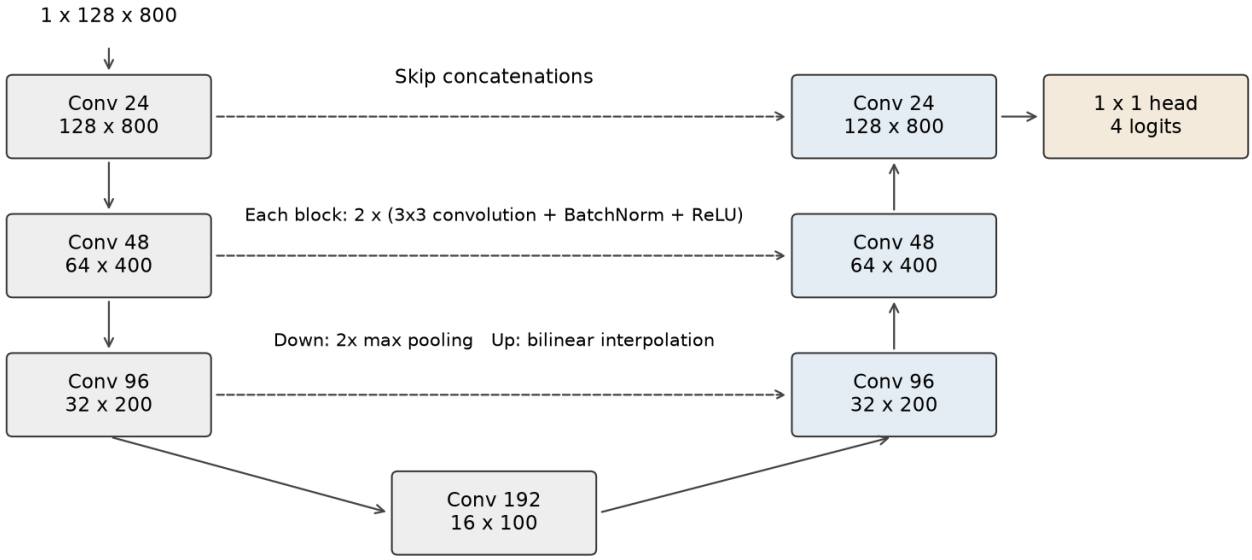
Preprocessing and input representation

RLE is decoded from one-based start/length pairs in column-major order. Image IDs, rather than annotation rows, define partitions. Final models use one grayscale channel with fixed (x - 0.5) / 0.25 normalization; masks retain four channels in A/B/C/D order. Joint flips preserve mask alignment, while brightness/contrast affect images only. The native strip geometry is retained.

The separate Milestone 2 exercise used the requested RGB float32 tensors in [0,1] and five training epochs. Its best checkpoint was epoch 4. Those exercise losses are not substituted for the later grayscale-model experiment.

Model architectures

Scratch U-Net. All weights start from random initialization. Encoder widths are 24, 48, 96 and 192, with three pooling stages. Decoder blocks concatenate matching encoder features to restore spatial detail. Each block has two 3x3 convolutions, BatchNorm and ReLU. Bilinear upsampling and a 1x1 four-channel head preserve the original mask size. The compact design follows U-Net's encoder-decoder structure and skip connections [1].



ResNet34 U-Net. An ImageNet-initialized residual encoder supplies multiscale features [2]; a U-Net decoder restores resolution. All encoder and decoder parameters are fine-tuned. The segmentation library adapts the first convolution to one grayscale channel by combining its RGB weights [5].

EfficientNet-B0 FPN. An ImageNet-initialized EfficientNet-B0 encoder [3] feeds a feature pyramid decoder with lateral multiscale fusion [4]. Pyramid channels are 128 and segmentation channels are 64. This segmentation adaptation provides a distinct decoder family and a smaller pretrained backbone. Encoder and decoder are both fine-tuned.

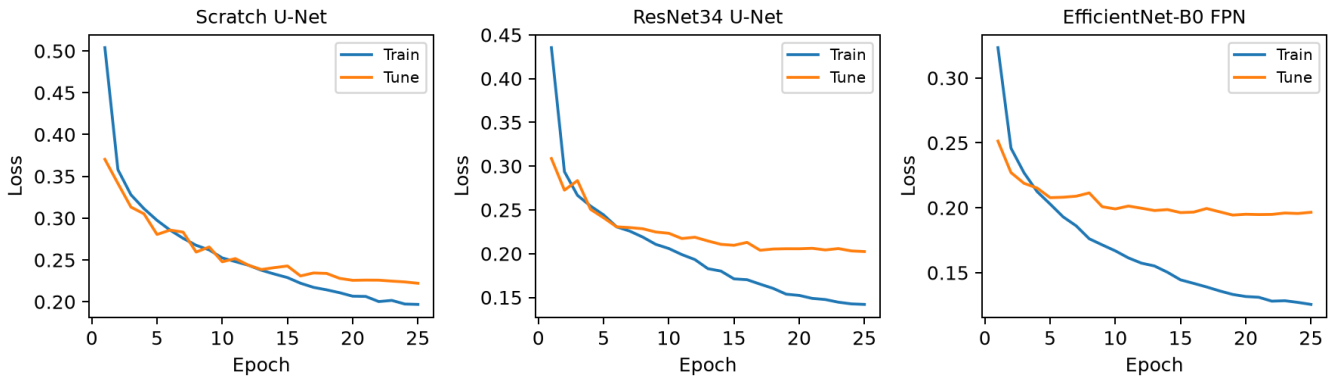
Four independent logits use sigmoid probabilities and a binary loss. The ground-truth classes do not overlap at pixels, but sigmoid predictions do not enforce exclusivity. A five-class softmax with background is a reasonable alternative to test. External competition solutions or extra labelled datasets were not used; ImageNet initialization supplies the required pretrained models.

Training protocol and experiment tracking

Seed 42 defines the original 80/20 image split, stratified by the complete label combination. The 20% holdout is split into 1,139 tuning and 1,140 audit images. Training uses 9,114 image IDs. Earlier milestone work also examined the outer holdout. The audit is excluded from current checkpoint and postprocessing choices, but is not a completely untouched project-level evaluation set.

Each model uses AdamW, weight decay 0.0001 and batch size 16, for at most 25 epochs. Initial learning rates are 0.001 for scratch and 0.0003 for pretrained models. A cosine schedule, gradient clipping at norm 1 and CUDA mixed precision control training. The lowest tuning-loss checkpoint is retained, with patience 7 for early stopping. A tempered inverse-prevalence sampler increases rare-positive exposure without forcing completely uniform classes.

The loss is 0.5 binary cross-entropy plus 0.5 Dice loss over positive image/category targets. BCE penalizes incorrect background predictions; positive-pair Dice emphasizes overlap without allowing abundant empty masks to dominate that term. Loss arithmetic uses float32. Real epoch measurements are logged to W&B; offline Kaggle logs are synced to the registered project after execution.



Metrics and selection

Per-class foreground Dice averages overlap on positive targets. Empty accuracy measures exact empty predictions on negative pairs. The local assessment combines 0.70 foreground Dice, 0.20 false-positive pixel score and 0.10 empty accuracy, averaged across classes. Pixel macro F1 and pixel accuracy are common secondary metrics. The actual Kaggle metric follows the same published component weights, but its internal aggregation is not assumed identical to this local implementation [6].

The selected architecture has the highest tuning assessment score. Thresholds {0.35, 0.50, 0.65} and minimum component areas {0, 32, 128} are a small per-class validation ablation. The public leaderboard is not used to choose these settings.

Measured model comparison

Model	Params M	Best epoch	Tune score	Audit score	Audit pixel F1	Audit pixel acc.
Scratch U-Net	1.096	25	0.6440	0.6322	0.3623	0.9916
ResNet34 U-Net	24.431	25	0.6818	0.6510	0.4759	0.9948
EfficientNet-B0 FPN	4.478	19	0.6872	0.6876	0.4979	0.9951

Inference ablation and cost

Variant	Forward passes	Tune score	Audit score	Audit pixel F1
single model	1	0.6872	0.6876	0.4979
horizontal-flip TTA	2	0.6966	0.6951	0.5266
equal-weight ensemble	3	0.7076	0.6839	0.5183

The three prespecified variants are a selected single model, its horizontal-flip TTA, and an equal mean of all three models. The rule prefers fewer forward passes when within 0.002 of the best tuning score. It selected **equal-weight ensemble**. Audit scores are reported after this choice and do not select the variant. Ensembling weights were not searched.

Horizontal-flip TTA has a higher audit score than the selected ensemble, despite a lower tuning score. The choice was kept frozen. This reversal illustrates uncertainty from a finite tuning split and the additional cost of three-model inference.

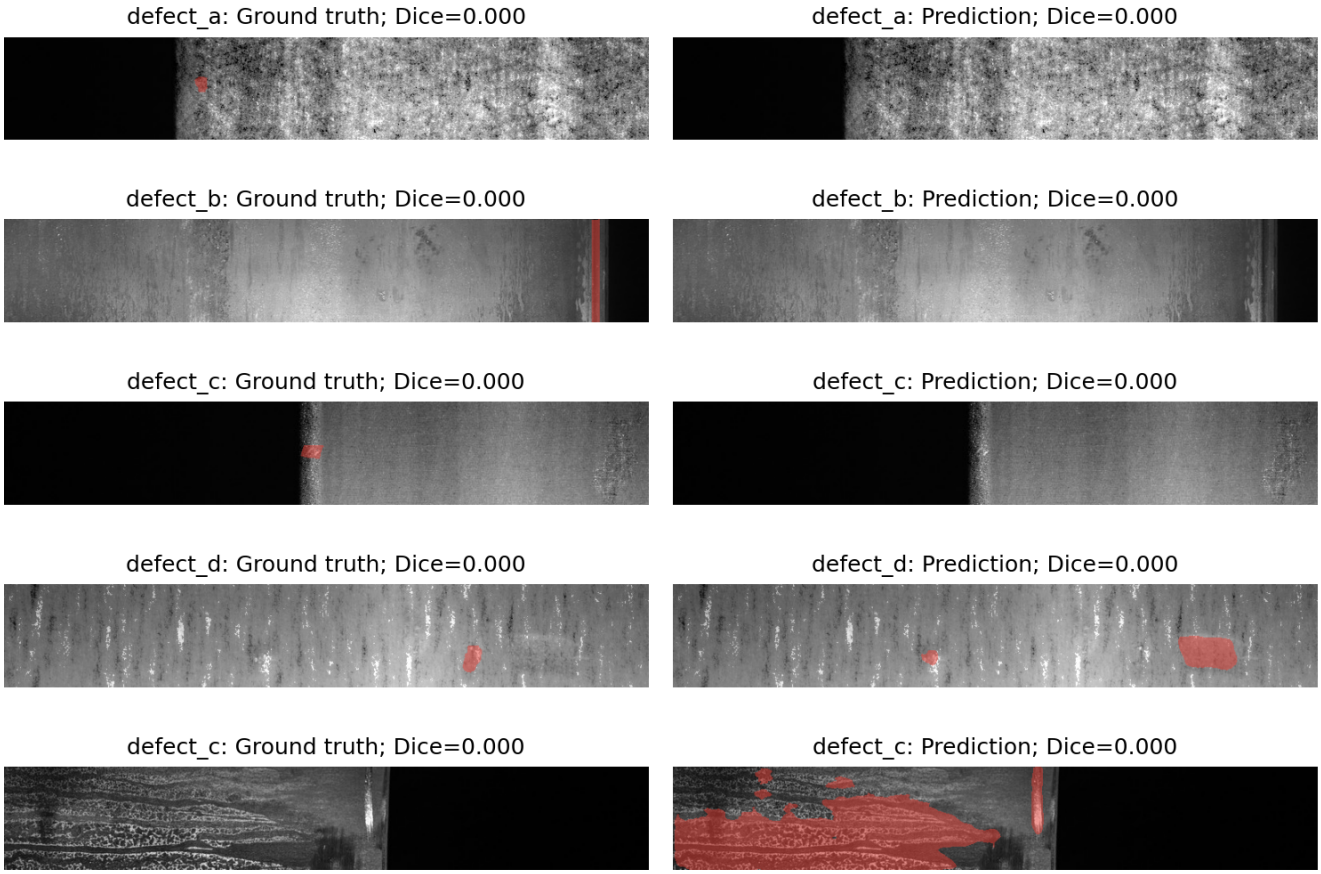
Selected variant by defect category

Class	Positive audit pairs	Threshold	Min area	FG Dice	Empty acc.
defect_a	108	0.35	32	0.4696	0.7161
defect_b	30	0.35	128	0.4162	0.9261
defect_c	410	0.35	128	0.6642	0.6808
defect_d	171	0.35	128	0.7612	0.8669

Small connected components can be noise, but removing them can also erase thin or tiny real defects. The selected area limits are therefore tied to validation evidence. Rare-class estimates have higher sampling uncertainty. Pixel accuracy should be read with foreground Dice and false-positive behavior, not as a stand-alone quality claim.

Audit error analysis

The examples below are selected from actual audit predictions: the lowest-Dice positive case for each category, plus the largest false-positive negative pair when present. Red overlays mark ground truth on the left and prediction on the right. They are diagnostic failures, not hand-picked demonstrations of success.



Across the selected audit predictions, 51 of 719 positive pairs have no predicted pixels, while 737 of 3,841 negative pairs contain at least one predicted pixel. These distinguish missed defects from false alarms. Weak boundaries, low contrast and small regions should be examined alongside class-level scores; the figure does not by itself prove a causal explanation for each failure.

Kaggle execution and deployment

The registered private notebook generated and submitted its own output CSV. Its public score is 0.67008; submission ID 57800905. Empty RLE cells use a space and row order matches the sample. The free static demo runs exported ONNX models in the browser. It was tested with native-size inputs and rejects incompatible dimensions. It returns four category areas and an overlay: https://huggingface.co/spaces/sarthak-wiz01/surface-defects-t32026-23f3000839.

Conclusions and future work

The final result comes from Scratch U-Net, ResNet34 U-Net, EfficientNet-B0 FPN, using the equal-weight ensemble inference variant. The main workflow improvements over the early baseline are longer supervised training, explicit rare-positive exposure, pretrained representation comparison, and validation-based false-positive control. The first five-epoch grayscale baseline scored 0.31426 publicly; the final public score above is measured separately from the local audit.

Further work should test an independent acquisition- or batch-level split, examine calibration for the rarest defects, and compare explicit pixel-level exclusivity with independent sigmoid heads. Repeated inspection can overfit a holdout. More epochs or more model combinations are not automatically useful; additional experiments should address a specific observed failure and include their inference cost.

Reproducibility and project record

The private repository contains modular data/training/inference code, executed notebooks, split files, histories, threshold trials, error metrics and the report builder. Milestone branches are retained with actual commit dates. Checkpoints and the offline dependency wheel are supplied as private Kaggle assets. Requirements and the recorded execution environment support rerunning the workflow.

GitHub: srthkdev/dlgenai-surface-defects-t32026
Registered Kaggle notebook
W&B experiment project

References

[1] Ronneberger, Fischer and Brox. U-Net, 2015.

[2] He et al. Deep Residual Learning for Image Recognition, 2015.

[3] Tan and Le. EfficientNet, 2019.

[4] Lin et al. Feature Pyramid Networks for Object Detection, 2017.

[5] Segmentation Models PyTorch v0.5.0, encoder input adaptation.

[6] SurfaceGuard AI competition evaluation and submission format.

[7] IITM DL and GenAI project guidelines, T3 2026.